

Statistics for Spatio-Temporal Data

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Statistics for Spatio-Temporal Data

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Preface

Nothing puzzles me more than time and space; and yet nothing troubles me less . . .

These words, by the English essayist Charles Lamb in his 1810 letter to Thomas Manning, provide a concise summary of this book. Of course, he was not thinking of Statistics, nor Science, nor Statistical Science, rather the more ephemeral notion of space and time. But here, in the physical world, there are scientific questions to resolve and predictions to make, and understanding the effects of dependencies across time and space is a crucial part. Those dependencies are in there somewhere, and we too are puzzled by them.

Up until the mid-twentieth century, Statistics' response to this puzzle was often to ignore the complicated structure, or to find clever ways to remove it. Interestingly, Charles Lamb seemed to be in tune with this, as he finished his sentence by saying, "as I never think about them." As Statistics has progressed through the twentieth century and moved into the twenty-first, spatial and temporal statistical methodology has been incorporated more and more into the scientific models of our world and indeed of our universe. As technology has improved, Statistics has been given a Rosetta Stone to begin to unlock Science's mysteries, from the molecular to the global to the cosmological.

In the beginning, there were data. Then, there were theories, formed from data, and those theories rose and fell according their agreement with new data. Data are central, and they should be cared for accordingly. To do good Science, databases should be fully documented and algorithms ("black boxes") involved in creating them should be in the public domain.

However, data alone do not tell us all that much about our world. When we look at data, how do we know if what we are seeing is "signal" as opposed to "noise"? How do we compare two sources of data—what is the basis of a comparison?

Similarly, theories (or models) by themselves are not often the best descriptions of the real world. What can be said about processes on scales at which

there were no observations? What about uncertainties in parameters, or forcings, or interactions with other processes? Indeed, the key is the proper blending of such models and the data. Sometimes this might be done informally, for example, by taking a simulated field, simulated from a mathematical (say) model, and visually comparing it to a field of actual data. From the visual comparison, a deficiency in the model might be obvious, which might lead to a parameter adjustment, or even a new parameter. Then a *new* simulation might be implemented and a new visual comparison made, and so forth. This is one way to combine data and model, but it is not a very efficient way to deal with either. Indeed, the power of Statistical Science is that it provides several frameworks in which to combine data and model, in optimal ways, for the purpose of scientific inference. It might seem strange that there is not just one framework in which to carry out inferences, but even Statistics has its tribes. However, the element that is common to all is an attempt to partition variability and to quantify uncertainty.

In this book, we take the firm stand that the best paradigm (to date) in which to partition variability and quantify uncertainty is the *hierarchical statistical model*. Such a model explicitly acknowledges uncertainty in the data, different from that in the process and parameters, and then it accommodates the uncertainty in the process (and finally in parameters, if necessary). We have used color in key places in our exposition, to distinguish between the parts of the hierarchical model concerned with data (green), with processes (blue) and with parameters (purple).

Hierarchical thinking (i.e., hierarchical modeling) is intimately tied to conditional thinking (i.e., modeling with conditional probabilities). Indeed, it is our perspective that conditional thinking is the aforementioned Rosetta Stone: It allows us to separately partition the effects of measurement error and scales of variability below the resolution of our data, conditioned on the process at possibly some other scale. Similarly, conditional thinking allows us to model a spatio-temporal process as it actually *evolves* through time, as opposed to just accounting for its marginal dependencies. Equally important, it allows us to use spatio-temporal dependencies in errors and/or parameters as a proxy for unknown and unknowable processes. Its impact goes even deeper, in that it allows parameters themselves to be dependent on other processes or other sources of data. Finally, it makes clear that if some components of variability are not interesting for a particular question, then the end result should *not* look like the data. Any unwanted components (e.g., measurement error) are filtered out: What you see (data) is not always what you want to get (process).

Critically, in the presence of data, conditional thinking allows sensible trade-offs to be made between data availability, process complexity, and computational complexity. Combined with the ever-expanding computational tools of the twenty-first century, the hierarchical statistical framework provides the structure to tackle important questions in Energy, Climate, Environment, Food, Finance, and so on. This will be a century of massive (spatio-temporal) datasets

collected to answer Society's dominant questions. In this book, we are particularly interested in inference for Climate and the Environment, where processes at small spatio-temporal scales influence those at larger scales, and vice versa. The questions to be resolved are fundamental to sustaining our planet, they involve complex spatio-temporal phenomena, *and* they are inherently statistical.

Our lives are spent marching through a space–time continuum, but space is different from time. We can (and often do) visit the same place over and over but always at different, ordered time points. We can go north, south, east, and west, up and down, but only ever forward from the past to the present and into the future. Any study of a spatio-temporal phenomenon needs to respect this difference. In this book, we have proposed spatio-temporal statistical methodologies that align with the underlying science, and we have found that hierarchical thinking is a natural way to achieve this alignment.

In the pages that follow, we have deliberately tried to build a bridge between the twentieth and twenty-first centuries in our presentation of spatio-temporal statistics. There are strong and powerful traditions that have developed in the last few decades and, even if they are not hierarchical, they provide the tools and motivation that can be used in hierarchical thinking. In some cases, hierarchical approaches that may be appealing in principle may be out of our reach in terms of timely implementation.

The official ftp site associated with the book can be found at:

ftp://ftp.wiley.com/public/sci_tech_med/spatio_temporal_data

In addition to posting errata, we will post supplemental material that we hope will be helpful to the reader.

As mentioned above, we have questions to resolve and predictions to make, so let's get started

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Acknowledgments

There are many people to thank, but two are missing. I wish my parents were alive to read this and know how important they are to what I do. I finally have a good answer to Ray's question, "How's the book going?"

My children, Amie and Sean, were only just present for my last book project, and now, as young adults, they are happily cheering on the completion of this one. Everyone has a book in them, and I hope they write theirs.

My dear friends and sibs may not understand all that is on these pages, but they know how important it was for me to "write it down." I would like them to know how important they are to me.

My co-author Chris has been a generous colleague and friend, and I admire enormously his intelligence and erudition. Together, we have been able to write a book that is so much more than what was in each of us; I have learned much from him. The order of authors is simply determined by the alphabet.

I have been taught by co-authors, colleagues, and students, who have shaped the material in this book. Their contributions are there in many, many ways. Some deserve special mention, which can be found in the joint acknowledgment.

The Department of Statistics at The Ohio State University has provided a nurturing environment for the whole lifetime of this book project. As part of my role as Director of the Department's Program in Spatial Statistics and Environmental Statistics, I have relied on my Program Assistant, Terry England, in many ways. This book has been a major project we have shared, and her efforts (which are described in the joint acknowledgment) are deeply appreciated and warmly acknowledged. Paul Brower, our Department Administrative Manager, has been a great supporter of this book project and, more generally, of my attempts to "save the planet." The office staff, the computer-support personnel, and the custodial staff are all warmly thanked for making my working environment in Cockins Hall a place where things get done . . . with good cheer.

Finally, it is not lost on me that this book will appear exactly 20 years after an earlier book I authored on spatial data. A lot can happen in two decades . . .

My rather unorthodox training in Statistics and Meteorology would not have been possible without the tremendous support of my advisors, Tsing-Chang (Mike) Chen and Noel Cressie, both of whom set a great example of having mastered the art of teaching and the more elusive skill of mentoring. Mike Chen opened my eyes to the beauty of dynamics through his courses in Dynamic Meteorology and Geophysical Fluid Dynamics (in particular) and his patient one-on-one expositions; he is a master of explaining the “real” meaning behind the mathematics. In addition to teaching these subjects, he taught me the *philosophy of science* and to appreciate the importance of knowing the history of your field. Noel has been an inspirational mentor, collaborator, and friend. In addition to opening my eyes to the beauty of spatial statistics, he has, more than anyone, taught me to think like a statistician, and to stand up for the things in which I believe scientifically. At the time I was choosing an advisor in Statistics, some students warned me that I did not want to work with Noel because he was too “hard.” One of the best decisions I ever made professionally was to ignore that advice! If hard means having high expectations, and providing generous advice and support, then I guess he was. Such a mentor has turned into a good friend. Noel mentioned at the beginning of this project how it can be very trying on a friendship to write a book. To be sure, we’ve had some disagreements, but we have managed quite well and I would not trade this journey for any other professional endeavor. I have learned a tremendous amount from Noel in writing this book, and I believe our friendship is stronger today than it was when we started.

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My path to Statistics was certainly not direct. In fact, the first classes I had in the subject as an undergraduate did little to impress upon me its importance. However, along the way I have had the great fortune of having some fantastic teachers and collaborators. In particular, I would like to thank Peter Sherman and Rol Madden, for introducing me to the power of spectral analysis and its scientific interpretation; and Mark Kaiser, for introducing me to hierarchical modeling in the early 1990s through a wonderful experimental course at Iowa State University. Many other people have influenced me along the way, either through conversations or collaborations. In particular, knowing that I am sure to unintentionally leave someone off of this list, I would like to thank: Chris Anderson, Thomas Bengtsson, Jim Clark, Bob Dorazio, William Dunsmuir, Dave Higdon, Tim Hoar, Dave Larsen, Andy Moore, Doug Nychka, Nadia Pinardi, Yanyan Sheng, Jon Stroud, Joe Tribbia, and Jay Ver Hoef. Of course, there are many others who have influenced me with the quality of their research, their integrity, and through brief discussions at meetings.

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N. C.
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